

10.2 The global patterns of food supply and demand

This chapter will explain:

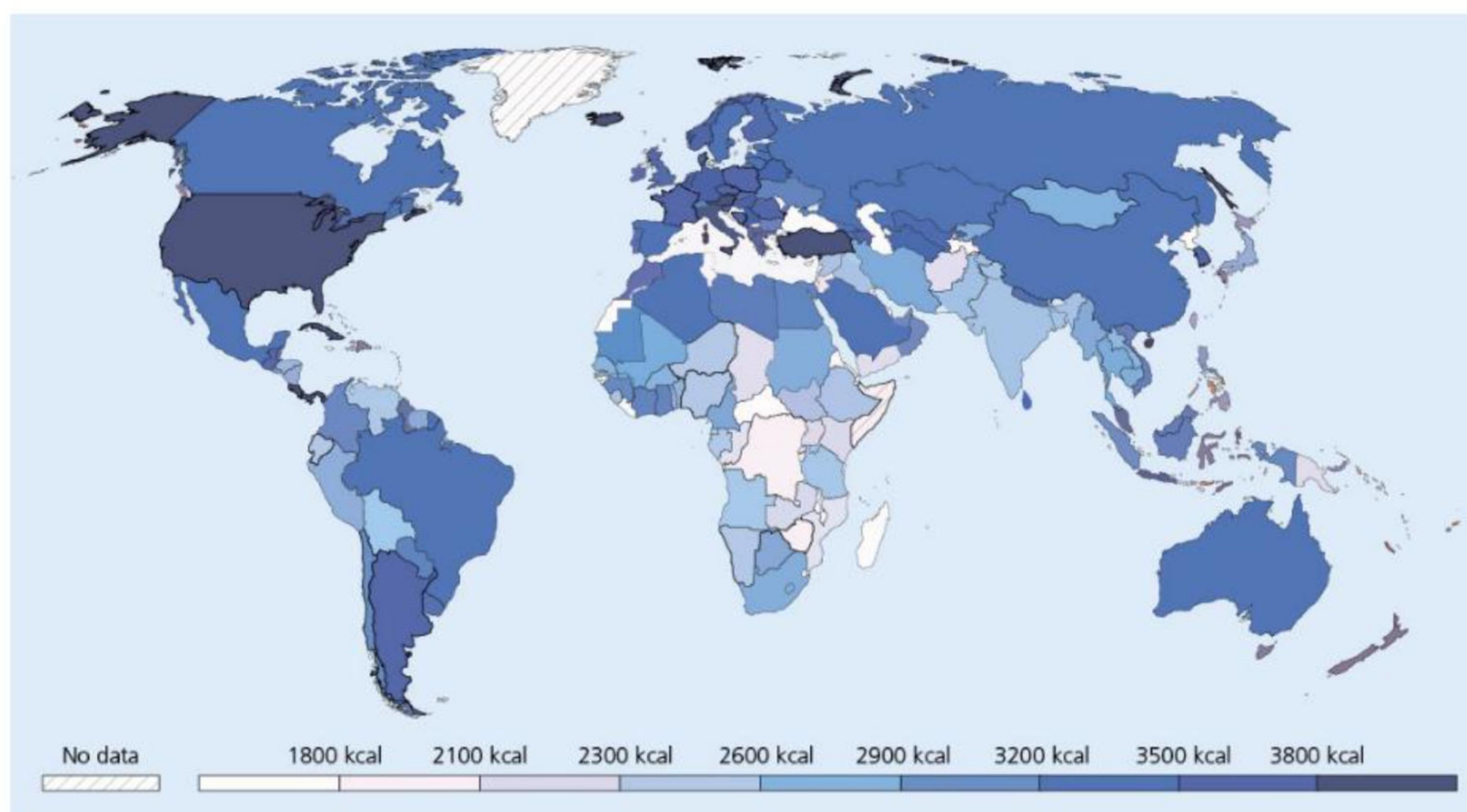
- ★ global variations in calorie intake and the reasons for this
- ★ the changing global production and consumption of food
- ★ the strategies used to increase food supply
- ★ the reasons for the globalisation of food supplies
- ★ the impacts of the globalisation of food supplies.

Global variations in calorie intake and the reasons for this

Per capita calorie supply is the main international indicator used to show variations in global access to food supply. It has limitations – a wholesome diet requires more than just energy and supply figures do not include consumption-level waste (food that is wasted at household, restaurant and retail levels). However, it is the best indicator available. [Figure 10.9](#) shows the global variation in the supply of calories for 2021, measured in kilocalories per person per day. The data comes from the Food and Agriculture Organization of the United Nations. It extends back to 1961 and is updated annually.

[Table 10.1](#) shows the changes by world region in per capita daily kilocalorie supply for 1980, 2000 and 2021.

- » Since 1960 there has been a consistent increase in global per capita calorie supply, but this has varied across the world's regions.
- » There has been a very significant increase in calorie supply in Africa and Asia in recent decades, resulting in the convergence of global trends in calorie supply.



▲ **Figure 10.9** The daily supply of calories per person, 2021

Variations in per capita global calorie supply have traditionally illustrated unequal access to food around the world. However, the strength of this relationship has weakened over time with the convergence of global trends in calorie supply. The relationship is still strong for lower-income and middle-income countries. [Table 10.1](#) shows clearly that Africa and Asia are still significantly below the world average, but to a much lesser extent than previously. The variations between the world regions that are above the world average have much less to do with calorie supply in terms of the lack of calories, but relate more to overeating.

▼ **Table 10.1** Per capita kilocalorie supply from all foods per day, 1980, 2000 and 2021

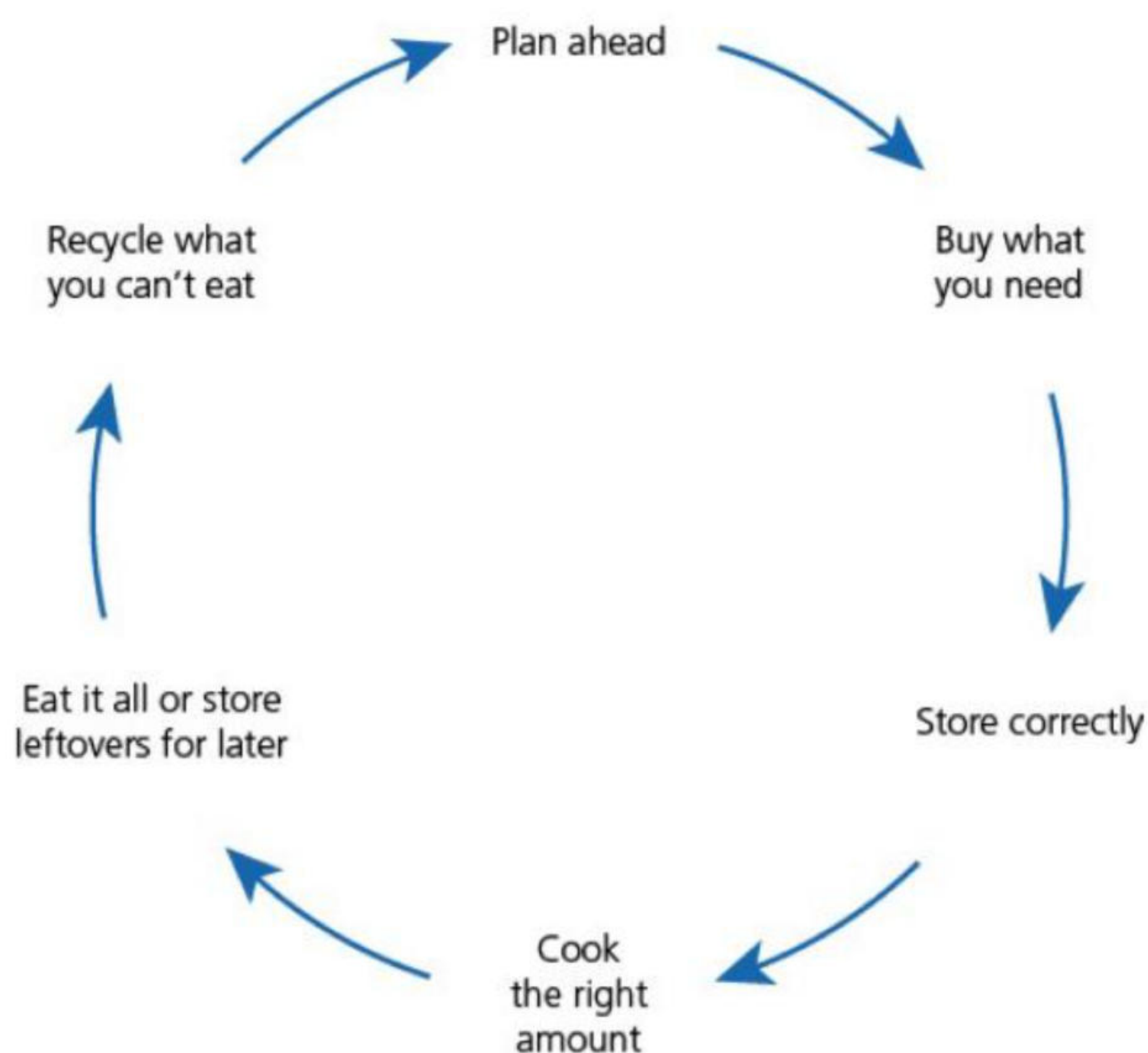
Region	1980	2000	2021
--------	------	------	------

World	2459	2669	2959
Europe	3348	3234	3456
North America	3102	3448	3578
Oceania	3013	3014	3087
South America	2658	2780	3108
Africa	2234	2405	2573
Asia	2186	2579	2927

Obesity and food waste

While the increase in global per capita calorie supply is to be welcomed for reducing levels of malnutrition in less affluent countries in particular, over 1 billion people globally are now classed as obese. A major cause of obesity is the increasing consumption of ultra-processed foods. The health crisis that this trend is creating is not confined to HICs; it is also affecting MICs and some LICs to an increasing degree. Many more people worldwide than ever before are regularly eating an unhealthy **diet**.

There has also been an increasing focus on identifying the amount of food that is wasted and what can be done to reduce this problem. [Figure 10.10](#) shows how individual households can reduce **food waste**. For many years, HICs have recorded high levels of food waste because people have bought more food than they need. However, in recent years, consumers in HICs in particular have become more aware of the issue of food waste, and as a result consumer behaviour is beginning to change. For example, there have been campaigns to get supermarkets to stop offers of 'buy one, get one free' and instead to reduce the price of individual items.



▲ **Figure 10.10** Methods of reducing food waste

Activities

- 1 Study [Table 10.1](#). Describe the main changes illustrated by the table.
- 2 Discuss what individual households can do to reduce food waste.

The changing global production and consumption of food

There is more food produced today than ever before, and more is being traded across international borders. Trade in food is a major aspect of globalisation and it is heavily dominated by a decreasing number of huge transnational corporations, which wield enormous power in the global food chain. The current global food system is the product of historical development. Food production has increased significantly since 1950 due primarily to considerable increases in:

- » world population, which has risen from 3 billion in 1960 to over 8 billion today
- » global wealth, particularly in emerging economies with a rapidly increasing 'middle class'

- » urbanisation of the global population – with urbanisation and associated rising incomes, households often eat greater and more diverse quantities of food.

These factors have driven a substantial increase in the demand for food, along with a change in dietary preferences towards more resource-intensive foods, particularly meat. The growth of Asia's economies, especially China and India, has created new consumer markets as a growing number of people who can be described as middle class demand better quality and more varied food. The demand for certain specialist types of food, for example organic and halal foods, have become clearly recognisable trends.

A small number of very large companies dominate the global food chain. For example, two companies control 40 per cent of the global commercial seed market, a much higher concentration compared to the year 2000. Other major aspects of the global food industry have become similarly concentrated. This is a trend aided by the increasing use of '**big data**' and artificial intelligence. Many people view this high level of control by a relatively small number of companies as worrying and anti-democratic.

Agro-industrialisation, or industrial agriculture, is the form of modern farming that refers to the industrialised production of livestock, poultry, fish and crops. This type of large-scale, capital-intensive farming originally developed in HICs, and has spread to MICs and some LICs since the beginning of the Green Revolution in the late 1960s ([page 254](#)). The characteristics of agro-industrialisation include:

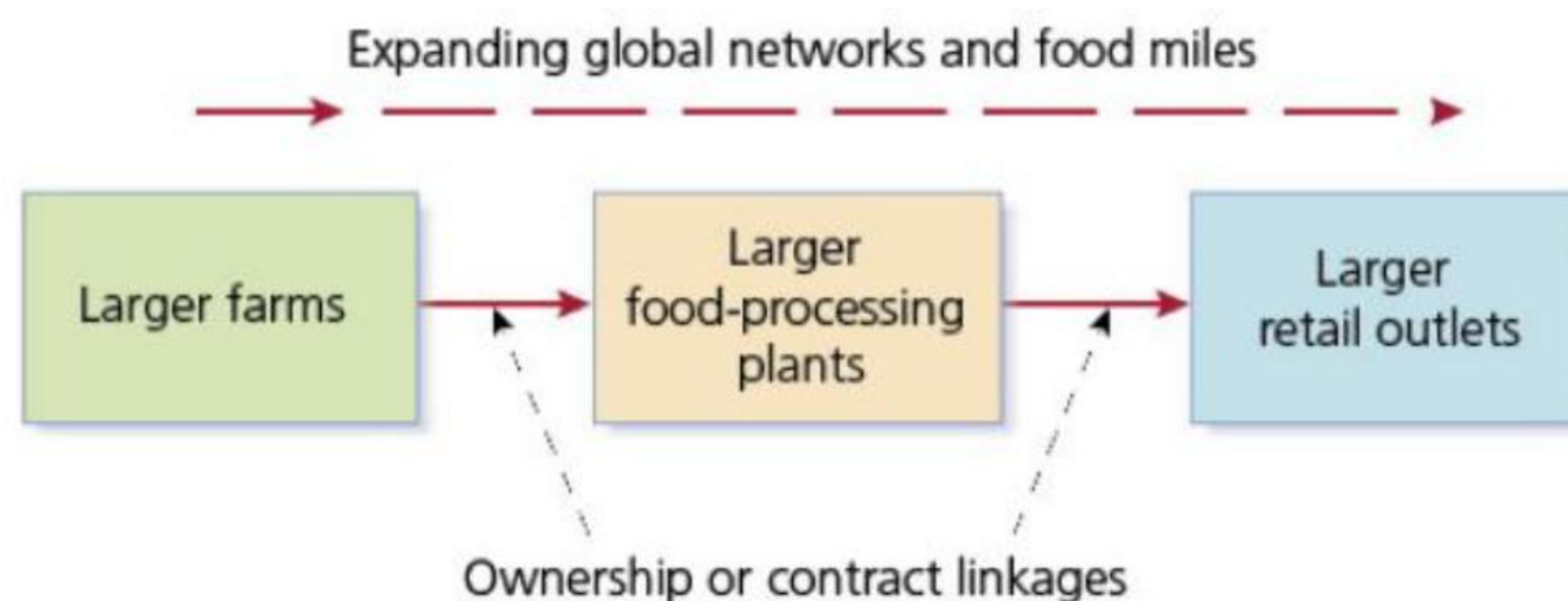
- » very large farms
- » concentration on one or a small number of farm products
- » a high level of mechanisation
- » low labour input per unit of production
- » heavy use of fertilisers, pesticides and herbicides
- » sophisticated ICT management systems
- » highly qualified managers
- » often owned by large agribusiness companies
- » often vertically integrated with food processing and retailing.

Not all farms and regions where agro-industrialisation is important will display all of these characteristics. For example, intensive market gardens may be relatively small, although the capital inputs are extremely high. Industrial agriculture is heavily dependent on oil for every stage of its operation. The most obvious examples of the use of oil are fuelling farm machinery, transporting produce and producing fertilisers and other farm inputs.

Agro-industrialisation is a consequence of the globalisation of agriculture, the profit ambitions of large agribusiness companies and the drive for cheaper food production. Vertical integration has become an increasingly important process, with growing linkages between the different stages of the food industry ([Figure 10.11](#)).

Regions where agro-industrialisation is clearly evident on a large scale include:

- » the corn and wheat belts in the USA
- » the Paris basin in France
- » the Pampas in Argentina
- » the Mato Grosso in Brazil.

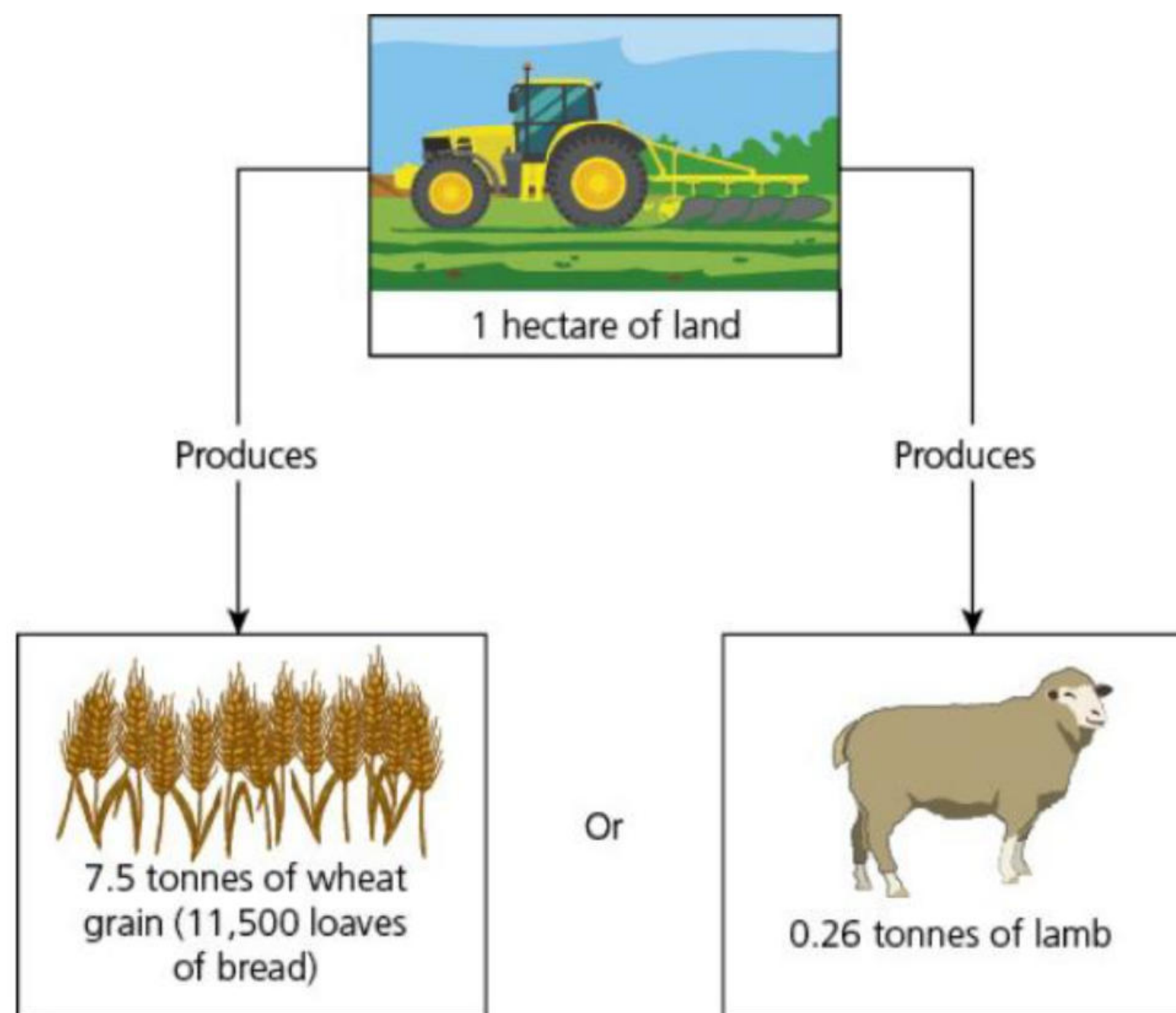


▲ **Figure 10.11** Agro-industrialisation: increasing vertical integration

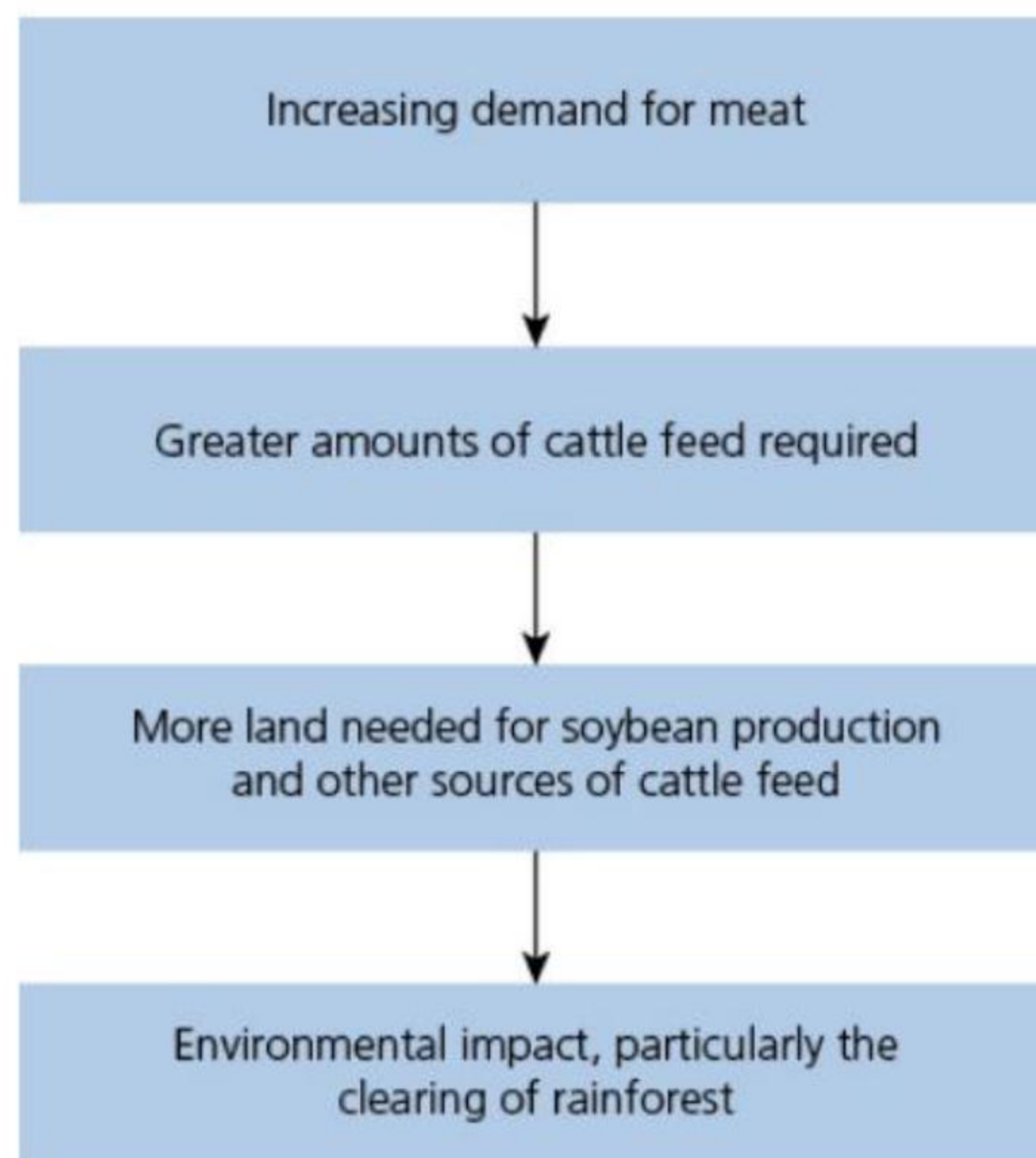
Food choices

High levels of meat consumption have been the norm in HICs for some time. However, as average incomes have increased in emerging economies in particular, the global demand for meat has risen rapidly. This is placing much greater demands on global food production, resulting in soil degradation and other environmental problems in various parts of the world. [Figure 10.12](#) compares the efficiency of energy conversion in arable and pastoral farming, while [Figure 10.13](#) looks at the environmental impact of the increasing demand for meat. The average efficiency of livestock at converting plant feed to meat is less than 3 per cent. Agricultural practice is not necessarily at fault, but maybe our choice of food is.

However, it should, of course, be remembered that livestock can produce more than just meat. Milk production is an important aspect of farming in many countries, and animals may also provide other valuable products such as wool and hides. In LICs, livestock may also provide an important source of labour.



▲ **Figure 10.12** A comparison of the efficiency of energy conversion in arable and pastoral farming



▲ **Figure 10.13** The environmental impact of the increasing demand for meat

Activities

- 1 State the three main reasons for the increase in global food production and consumption.
- 2 **a** Define agro-industrialisation.
b Give three characteristics of agro-industrialisation.
c State three examples of regions where large-scale agro-industrialisation occurs.
- 3 Explain [Figure 10.12](#).

The strategies used to increase food supply

The Green Revolution

The package of agricultural improvements generally known as the **Green Revolution** was seen as the answer to the food problem in many parts of the developing world in the post-1960 period. India was one of the first countries to benefit, when its High Yielding Variety Programme (HYVP) started in 1966–67. In terms of production this was a turning point for Indian agriculture, which had virtually reached stagnation. The programme introduced new hybrid varieties of five cereals: wheat, rice, maize, sorghum and millet. All

were drought-resistant (with the exception of rice), were very responsive to the application of fertilisers, and had a shorter growing season than the traditional varieties they replaced ([Figure 10.14](#)). Although the benefits of the Green Revolution are clear, serious criticisms have also been made. The two sides of the story can be summarised as follows.



▲ **Figure 10.14** Green Revolution crops being harvested in Brazil

The advantages of the Green Revolution

- » Yields are two to four times greater than for traditional varieties.
- » The shorter growing season has allowed the introduction of an extra crop in some areas.
- » Farming incomes have increased, allowing the purchase of machinery, better seeds, fertilisers and pesticides.
- » The diet of rural communities is now more varied.
- » Local infrastructure has been upgraded to accommodate a stronger market approach.
- » Employment has been created in industries supplying farms with inputs.
- » Higher returns have justified a significant increase in irrigation.

The disadvantages of the Green Revolution

- » High inputs of fertiliser and pesticide are required to optimise production. This is costly in both economic and environmental terms. Rural indebtedness has risen sharply in some areas.
- » HYVs require more weed control and are often more susceptible to pests and diseases.
- » Middle- and higher-income farmers have often benefited much more than the majority on low incomes, thus widening the income gap in rural communities. Increased rural-to-urban migration has often been the result.
- » Mechanisation has increased rural unemployment.
- » Some HYVs have an inferior taste.
- » The problem of salinisation has increased along with the expansion of irrigated areas.

In recent years a much greater concern has arisen about Green Revolution agriculture. The problem is that the high-yielding varieties introduced during the Green Revolution are usually low in minerals and vitamins. Because the new crops have displaced the local fruits, vegetables and legumes that traditionally supplied important vitamins and minerals, the diet of many people in the developing world is now extremely low in zinc, iron, vitamin A and other micronutrients.

The Green Revolution has been a major factor in enabling global food supply to keep pace with population growth, but with growing concerns about a new food crisis, new technological advances may well be required to improve the global food security situation.

The New Green Revolution

Agricultural economists have talked about the concept of a **New Green Revolution** in recent decades, which draws upon the best of the agricultural technologies currently available. It emphasises improved farm management in order to minimise environmental damage from external inputs and aims to benefit low-income farmers and marginal areas bypassed by the original Green Revolution. Examples of this approach include:

- » breeding crop varieties that can cope with adverse conditions, such as salt-tolerant rice and more drought-resistant sorghums and millets
- » seeds produced to improve the nutritional value of crops as well as to provide more calories
- » soil nutrient cycling through crop rotation and biomass recycling
- » reliance on genetic resistance to pests and disease to replace chemical and mechanical pest control.

This new process aims to blend farmers' traditional techniques with new knowledge in crop and animal husbandry. For example, integrated pest management (IPM) uses natural predators such as spiders and

wasps to fight pests, reducing both costs to farmers and environmental damage.

Activities

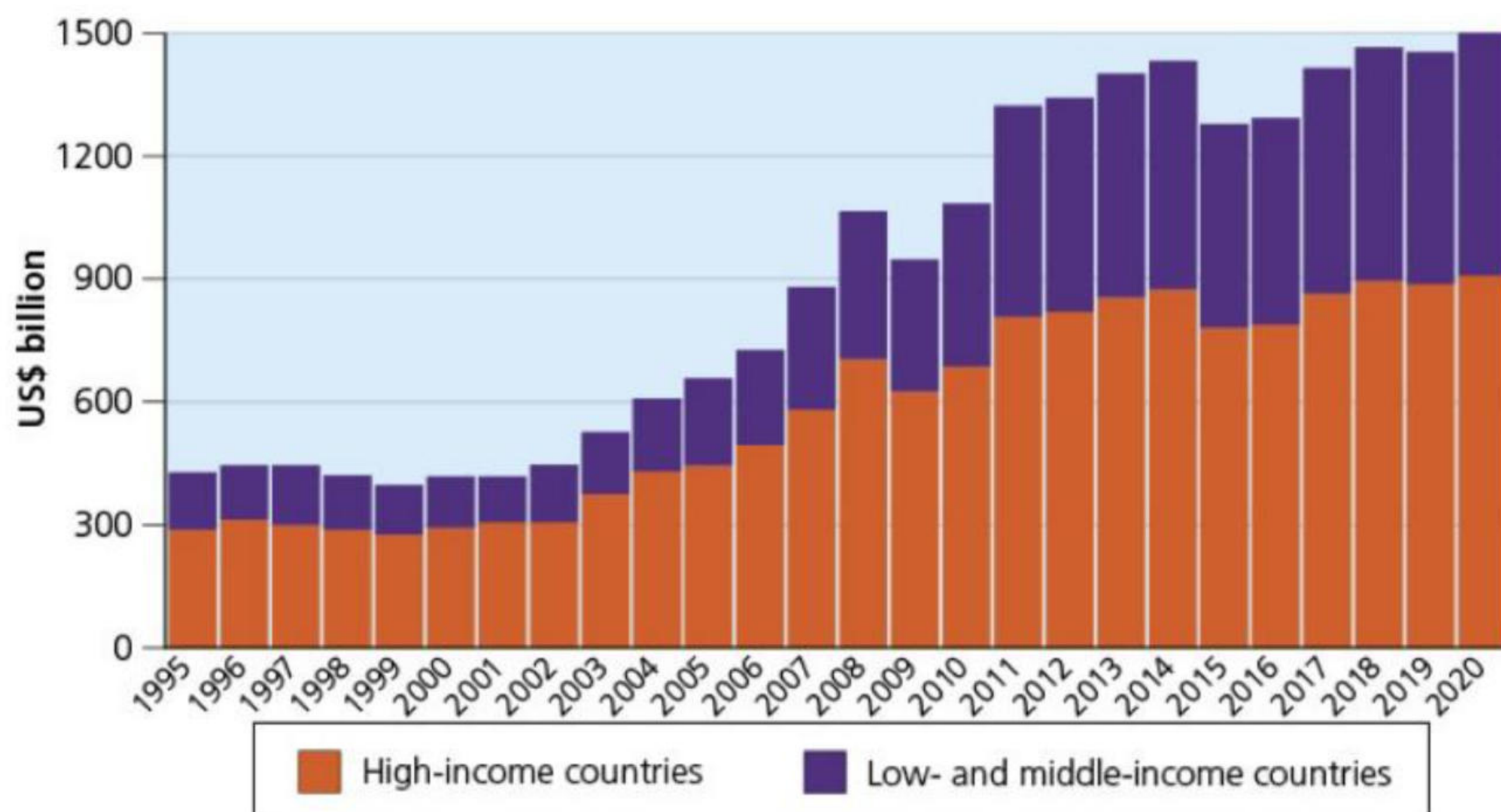
- 1 Define the Green Revolution.
- 2 Discuss three advantages and three disadvantages of Green Revolution farming.
- 3 In what ways is the New Green Revolution introducing new farming practices?

The reasons for the globalisation of food supplies

Reducing the barriers to trade in agricultural products

While trade in manufactured goods was gradually liberalised (made easier by reducing the barriers to trade – tariffs, quotas, regulations) after the General Agreement on Tariffs and Trade in 1947, it was not until the World Trade Organization Agreement on Agriculture in 1995 that agriculture as a sector was explicitly included in trade liberalisation at the multilateral level. As [Figure 10.15](#) shows, trade in food and agricultural products increased significantly after 2000.

- » Brazil, China and other emerging economies increased their share of global trade in food and agricultural products.
- » The share of global exports originating in LICs and MICs increased from about 30 per cent in 1995 to 40 per cent in 2011, and since then has remained fairly constant.
- » HICs account for about 60 per cent of food and agricultural exports.



▲ **Figure 10.15** Global food and agricultural trade, 1995–2020

The increase in food and agriculture trade since 2000 was also due to increased connectivity between countries, with more countries expanding their participation in global food and agricultural trade. Today countries are connected to more trade partners than ever before in terms of food and agriculture.

The increased efficiency of seaborne transport

Seaborne shipping is by far the most important mode of transport in the international trade in agricultural products. An increasing part of this trade is carried in containers. The majority of world containers are moved through liner shipping services, which are regular transport services provided by global shipping companies. These services provide a dense network that connects ports and countries around the world. The improved logistics of seaborne shipping and of **intermodal transport hubs** has been beneficial to the international trade in food and agricultural products.

The rising demand for international food products

Tourism and global marketing have introduced people in an increasing number of countries to a much wider range of global foods than previous generations had. This is reflected in the eclectic range of international foods stocked in large supermarkets in many countries. Major restaurant and takeaway companies such as KFC and McDonald's have expanded their global networks in terms of both ownership and franchising. Certain countries and cultures are internationally known for their food – the evidence can be seen in restaurants with national identities. Italian, French, Mexican, Chinese, Indian and Thai restaurants are associated with a particular country and are also frequently found in other countries. As a class, you may be able to think of the particular foods that are associated with such restaurants.

Interesting notes

- ★ The largest food-exporting countries are the USA, the Netherlands and China.
 - ★ The largest net exporters, which export more than they import, are Brazil, Argentina and Spain.
 - ★ The largest net importing countries are China, Japan and the UK.
-

Activity

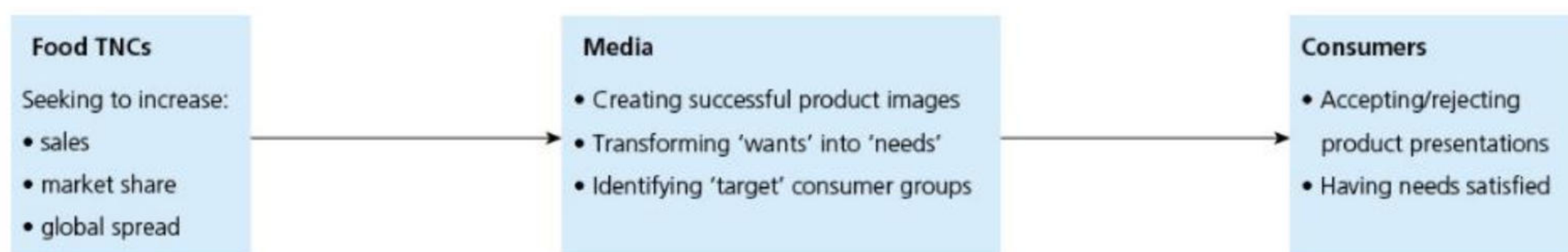
Study [Figure 10.15](#). Describe and explain the changes illustrated by the graph.

The impacts of the globalisation of food supplies

Globalisation has had profound effects on the world food system. As major transnational companies have increased their ownership and control of food production, distribution and consumption, more food products have crossed international borders. A wider range of foods has become available to consumers around the world. In many countries this has resulted in the year-round availability of seasonal and non-native foods.

The globalisation of the world food system pushed forward major changes in food production practices, with a big shift towards monoculture and larger-scale farming. While these changes have improved productivity and lowered costs, there has been an increasing negative impact on environments around the world. The impact on local agricultural systems has been overwhelming in many countries and regions. Here, the historic cultivation of local foods for the resident population has frequently given way to production for export markets. Such a massive change can leave communities vulnerable to fluctuations in global food prices and dependent on imported foods.

The power and influence of global food brands ([Figure 10.16](#)) has led the way in the expansion of Western dietary patterns. This has impacted both on cultural food heritage and on health in terms of obesity and other forms of ill-health associated with a significant consumption of ultra-processed food.



▲ **Figure 10.16** Food TNCs, the media and consumers

The carbon emissions associated with larger cross-border movements of food have increasingly contributed to climate change. The term 'food miles' was first used in the 1990s to draw attention to this expanding environmental impact. **Food miles** can be defined as the distance that food travels from the farm where it is produced to the plate of the final consumer. It is an indication of the environmental impact of food consumption. However, it is important to remember that many other processes also contribute to the carbon footprint of food, including:

- » farming methods
- » processing and packaging
- » refrigeration.

The intensification of agricultural practices brought about by the Green Revolution in MICs and LICs, along with a range of other advances that characterise modern agriculture, has resulted in serious negative consequences, particularly:

- » ecological degradation
- » unsustainable resource consumption
- » dependency on non-renewable resources such as fossil fuels.

Activities

- 1 How has globalisation impacted traditional agricultural production in many parts of the world?
 - 2 **a** Define food miles.
b Why are food miles an important environmental concept?
-